Atoms – the source of all our electricity

(This unit should be attempted after the atomic structure chapter in Material World.)

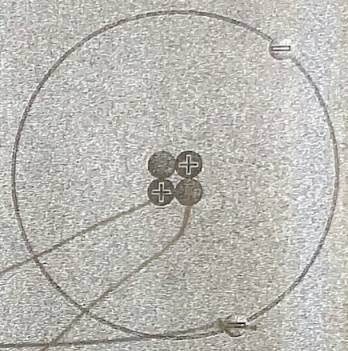
To understand where electricity comes from, you need to know a little about the structure of atoms.

Atoms are made up of three kinds of subatomic particle:

- **Protons** and **neutrons** in the central nucleus.
- **Electrons**, in orbits around the nucleus.

Protons carry a positive electric charge and electrons carry a negative charge.

Neutrons have no charge – they are electrically neutral.



The structure of a helium atom

1. Draw label lines from the two bullet points to the matching parts of the atomic model in the preceding diagram.
2. Are electrons positive or are they negative? Negative
   How many electrons are there in a helium atom? 2
3. Are protons positive or are they negative? Positive
   How many protons are there in a helium atom? 2
4. What is the overall charge on a helium atom? 0
5. Now use the following word list to complete sentences a. to c. below.

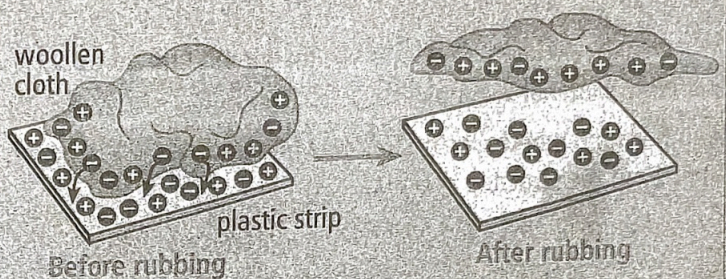
   everything neutral equal zero all electrons

   a. In the same way as the helium atom, every other atom contains equal numbers of (+ ve) protons and (– ve) electrons, so the overall charge on any atom is zero; all atoms are therefore electrically neutral.
   b. Electricity arises when electrons move from one atom to another.
   c. Since everything is made of atoms, it follows that all materials have the potential to produce electricity.



Rubbing atoms together

In the following diagram, a woollen cloth is shown being rubbed against a plastic strip. Friction causes some electrons to jump from the cloth to the plastic, as shown by the arrow in the diagram.



Model of electrons jumping from cloth to plastic strip

1. Count the numbers of (+) ve and (–) ve charges after rubbing, on each material in the diagram above, and complete the following table.

	Woollen cloth	*Plastic strip*
Charges before rubbing	+6 −6 = 0	+6 −6 = 0
Charges after rubbing	+6 − 3 = 3	+6 − 9 = −3

2. Describe what happens to the positive charges (protons) during the rubbing.

They do not move

3. Use the following word list to complete sentences a–d below.

charge protons insulator conductor friction nucleus surface electrons

a. The rubbing of two materials together always causes friction.

b. Friction causes electrons to jump from the atoms on the surface of one material onto the atoms of the other material.

c. The protons and neutrons remain where they are, bound in the nucleus of each atom.

d. If one of the materials is a conductor, the charges quickly spread out through the material, but if one is an insulator, the charges are trapped on the surface. The material is then said to possess a charge.

4. Turn back to the description of the structure of an atom to explain why only electrons can be moved from an atom, and not protons.

The protons are in the nucleus of the atom

Atoms – the source of all our electricity

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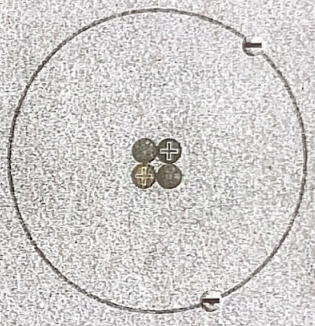
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Neutrons have no charge – they are electrically neutral.



The structure of a helium atom

1. Draw label lines from the two bullet points to the matching parts of the atomic model in the preceding diagram.
2. Are electrons positive or are they negative? Negitive

   How many electrons are there in a helium atom? TWO
3. Are protons positive or are they negative? Positive

   How many protons are there in a helium atom? TWO
4. What is the overall charge on a helium atom? none /0
5. Now use the following word list to complete sentences a. to c. below.

   ~~everything~~ ~~neutral~~ ~~equal~~ ~~zero~~ ~~all~~ ~~electrons~~

   a. In the same way as the helium atom, every other atom contains equal numbers of (+ ve) protons and (– ve) electrons, so the overall charge on any atom is zero; all atoms are therefore electrically Nuteral.

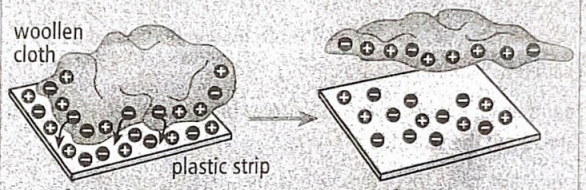
   b. Electricity arises when electons move from one atom to another.

   c. Since everything is made of atoms, it follows that all materials have the potential to produce electricity.



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2. Describe what happens to the positive charges (protons) during the rubbing.

They do not move

3. Use the following word list to complete sentences a–d below.

charge protons insulator conductor friction nucleus surface electrons

a. The rubbing of two materials together always causes friction.

b. Friction causes electrons to jump from the atoms on the surface of one material onto the atoms of the other material.

c. The protons and neutrons remain where they are, bound in the nucleus of each atom.

d. If one of the materials is a conductor, the charges quickly spread out through the material, but if one is an insulator, the charges are trapped on the surface. The material is then said to possess a charge.

4. Turn back to the description of the structure of an atom to explain why only electrons can be moved from an atom, and not protons.

The electrons are in the outer parts of the atom.

Atoms – the source of all our electricity

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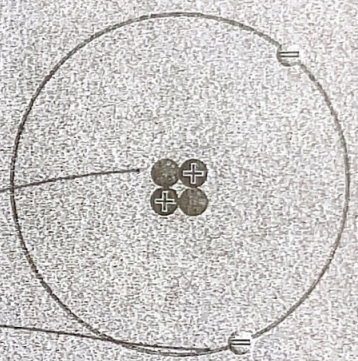
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Neutrons have no charge – they are electrically neutral.



The structure of a helium atom

1. Draw label lines from the two bullet points to the matching parts of the atomic model in the preceding diagram.
2. Are electrons positive or are they negative? ~~Positive~~ Negative
   How many electrons are there in a helium atom? 2
3. Are protons positive or are they negative? Positive
   How many protons are there in a helium atom? 2
4. What is the overall charge on a helium atom? 0
5. Now use the following word list to complete sentences a. to c. below.

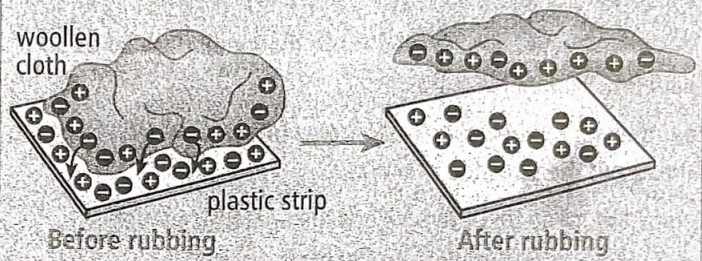
   everything neutral equal zero all electrons

   a. In the same way as the helium atom, every other atom contains equal numbers of (+ ve) protons and (– ve) electrons, so the overall charge on any atom is zero; all atoms are therefore electrically neutral
   b. Electricity arises when electrons move from one atom to another.
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2. Describe what happens to the positive charges (protons) during the rubbing.

nothing

3. Use the following word list to complete sentences a–d below.

charge protons insulator conductor friction nucleus surface electrons

a. The rubbing of two materials together always causes friction.

b. Friction causes electrons to jump from the atoms on the surface of one material onto the atoms of the other material.

c. The protons and neutrons remain where they are, bound in the nucleus of each atom.

d. If one of the materials is a conductor, the charges quickly spread out through the material, but if one is an insulator, the charges are trapped on the surface. The material is then said to possess a charge.

4. Turn back to the description of the structure of an atom to explain why only electrons can be moved from an atom, and not protons.

Because electrons are on the outside no trapped in the nucleus

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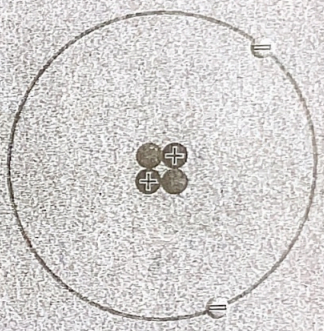
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The structure of a helium atom

1. Draw label lines from the two bullet points to the matching parts of the atomic model in the preceding diagram.
2. Are electrons positive or are they negative? ~~Negative~~ negative
   How many electrons are there in a helium atom? 2
3. Are protons positive or are they negative? ~~Positive~~ Positive
   How many protons are there in a helium atom? 2
4. What is the overall charge on a helium atom? ~~8~~ 0
5. Now use the following word list to complete sentences a. to c. below.

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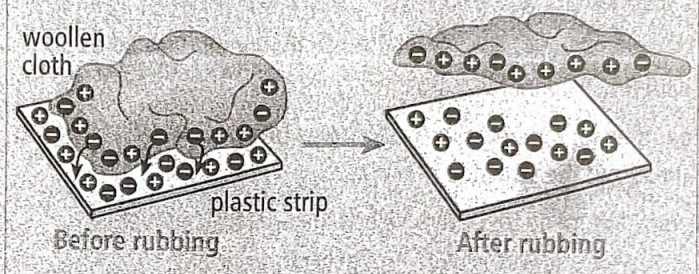
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They do not move

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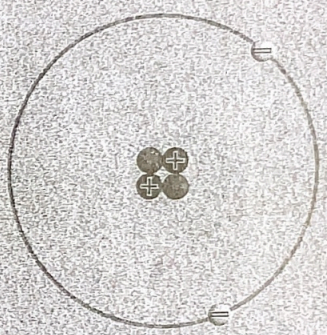
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3. Are protons positive or are they negative? Postive

   How many protons are there in a helium atom? 2
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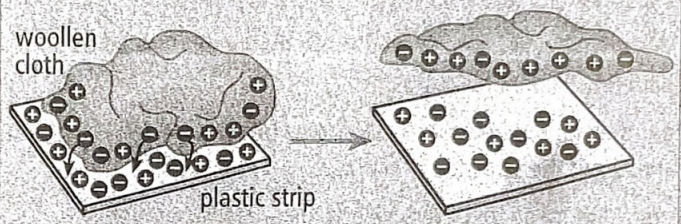
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